Observations

Lab note for Bramble hybrid fusion.

Keyword search and dense search were missing different passages.

The observations follow.

Bramble fusion with reciprocal rank fusion at k equals 60 beat either retriever alone on the held-out question set.

Dense search recovered paraphrases while keyword search recovered rare identifiers such as ticket codes.

The working decision is: Ship reciprocal rank fusion as the default desk search and keep the two raw lists available to reviewers..

Measurement

How it was measured: Each question was scored with keyword search alone, dense search alone, and fused ranks, using the same top-20 cutoff.

What came back: Fused recall at ten was 0.84 compared with 0.76 for dense search and 0.71 for keyword search.

What this does not show: The fusion trial did not tune k per domain, so the constant 60 may be suboptimal for very short notes.

Still unresolved: Would a learned gate outperform a fixed fusion constant on mixed-length collections? Operational risks: Severity medium: Fusion can promote a keyword-only hit that is lexically similar but topically wrong.

Recommended action: Show both retriever ranks beside the fused list during review..